BCA 5th Semester

MINOR PROJECT

Question Bank

1. During the planning phase of your project, how did you gather and document the requirements? Describe the process you followed to ensure that the technical requirements aligned with the client or stakeholder’s expectations.
2. Explain how you created a project roadmap for your technology-specific project. What tools or methodologies (e.g., Gantt charts, Agile sprint planning) did you use to outline project milestones, deliverables, and timelines?
3. In the context of your project, how did you assess the feasibility of different technology options? Describe how you selected the appropriate technology stack (e.g., programming language, frameworks, databases) for the project and the criteria you considered.
4. Describe the architecture of the project you worked on, focusing on the specific technologies used. How did the design of the architecture support the system’s goals, such as scalability, security, or performance?
5. Explain how you approached the design phase for your technology-specific project. What design patterns, frameworks, or principles (e.g., MVC, microservices, RESTful API design) did you apply, and how did they contribute to the project’s structure and efficiency?
6. Discuss how you addressed the trade-offs between different technologies during the design of the project (e.g., choosing between a monolithic vs. microservices architecture, SQL vs. NoSQL databases). How did these decisions impact the final implementation?

7. Explain the role of version control and collaboration tools (e.g., Git, GitHub, GitLab) in your project's development process. How did your team manage code reviews, feature integration, and resolving merge conflicts?
8. Provide a detailed account of how a specific feature was implemented in your project. Include the technologies you used (e.g., programming languages, frameworks, libraries) and how the feature integrated with the rest of the system.
9. Describe any issues or bugs encountered during the development process and how you resolved them. What debugging techniques or tools (e.g., IDEs, logging frameworks, debuggers) did you use to identify and fix these issues?
10. Discuss the integration of any third-party services, APIs, or external libraries in your project. How did you ensure compatibility, and what steps were taken to manage the dependencies between your project and the external systems?
11. Explain how the use of APIs or external services (e.g., payment gateways, cloud services, machine learning APIs) enhanced the functionality of your project. How did you handle challenges such as rate limits, authentication, or API versioning?
12. In your project, how did you ensure that security best practices were followed? Describe how you handled sensitive data, implemented authentication and authorization mechanisms, and secured communication (e.g., encryption, HTTPS).
13. Explain the data management strategy you employed in your project. How did you handle database design, optimization, and ensuring data integrity? Were any specific technologies (e.g., SQL databases, NoSQL solutions) used for managing data?

14. Discuss the testing strategy used for your project. How did you ensure that the code was thoroughly tested (e.g., unit tests, integration tests, system tests)? What testing frameworks or tools (e.g., JUnit, Selenium) were employed?

15. Describe how you implemented continuous integration and continuous deployment (CI/CD) pipelines in your project. How did automation of builds, tests, and deployments improve the quality and delivery of your project?
16. Explain the process of manual testing and user acceptance testing (UAT) in your project. How did you validate that the final product met the client's or stakeholder's expectations?
17. How did you ensure the performance of your project, particularly in terms of response times, load handling, and resource management? What performance testing or monitoring tools (e.g., JMeter, New Relic) did you use, and what were the results?
18. Describe any performance bottlenecks you encountered during the testing phase and how you optimized them. What techniques (e.g., caching, database indexing, code optimization) were used to improve the performance?
19. Explain the deployment process of your project. What technologies (e.g., Docker, Kubernetes, AWS, Azure) were used to deploy the project, and how did you ensure that the deployment was seamless and scalable?
20. After deployment, what monitoring or logging tools (e.g., ELK stack, Prometheus, Grafana) did you set up to ensure the system's health and performance? How did you address any post-deployment issues?